

BOAMP Digital Procurement Study

Successor Linkage, Survival, and Trend Methodology

BOAMP Data Science Internship Project

2026-08-15T18:58:06

Contents

1	Context	2
2	What The Pipeline Measures	2
3	End-to-End Pipeline	3
4	Grand Ouest Regional Reference	3
5	Linkage Algorithms	4
6	Survival Modeling	5
7	Trend And Change-Point Detection	6
8	Technology Segmentation	7
9	Dev Results	8
10	Internal Held-Out Reference Results	8
11	Quality Evidence And Interpretation	9
12	Model-Assisted Challenge Review	10
13	Modeling Tables	10
14	Survival Results	10
15	Trend Results	13
16	Defensible Decision	14
17	Limitations And Robustness	14
18	Perspectives	15
18.1	Causal Inference (Outline Only)	15
19	Recommended Reporting Position	15
20	Current Source Files	16
21	Methodological References	16

Technical Summary

This report states the current defensible BOAMP successor-linkage pipeline. The study does not claim to certify legal contract renewals. Its event is an *observable successor procurement*: a later BOAMP procurement episode from the same buyer that is sufficiently similar to an earlier awarded digital procurement episode.

The current Grand Ouest survival cohort contains 3,800 awarded digital procurement episodes. The main linkage rule accepts 544 successor links, giving an observed event rate of 0.143 and a median observed successor time of 31.82 months among linked events. The candidate pool is broad: 763,417 candidate pairs from 3,520 anchors with at least one candidate.

Linkage accuracy is read against the Grand Ouest regional reference: a stratified review of 120 anchors of which 88 are usable, carried out against real BOAMP notices before these methods existed. On its locked split of 72 anchors (18 with a reviewed successor), `M_B_text_ranking @ 0.70` has precision 0.875 (95% CI 0.529–0.978), recall 0.389 (95% CI 0.203–0.614), false-positive rate 0.000, and 8 accepted links. Recall is capped at 0.913 by candidate generation, which reaches 21 of the 23 reviewed successors. These are reference-sample estimates on fewer than a hundred anchors, not independently validated accuracy: the labels were generated by a single LLM research pass and spot-checked on a subset by the project owner, not verified anchor-by-anchor and not reviewed by an independent specialist panel. `M_C_weighted_gated` recovers more positives (recall 0.667) but admits more false positives (FPR 0.185). This trade-off is important because a false positive in a survival dataset creates both a false event and a false event time, while an abstained link is handled conservatively as right-censoring.

1 Context

Gigalis is the digital central purchasing body (*centrale d’achat numérique*) for public institutions in the Pays de la Loire region and beyond. It designs pooled framework agreements in cloud, cybersecurity, digital infrastructure, and artificial intelligence, and its strategic relevance depends on anticipating members’ future needs. The internship’s central question is: using historical public digital procurement data, how can the probability that a contract or technological segment generates an identifiable purchasing need within the next 12–24 months be estimated, and how can that estimate inform Gigalis’s central purchasing strategy?

This decomposes into three sub-problems: a *lifetime problem* (survival analysis: how long between a digital procurement episode and its successor?), a *trend problem* (change-point detection: which technological segments are growing, plateauing, or declining?), and a *text-signal problem* (NLP: extracting a contract’s technological theme automatically). This project answers the first two in depth; the NLP sub-problem is addressed with a reproducible coarse substitute rather than a trained classifier (§8).

BOAMP does not encode legal contract renewal as an explicit field, so the study cannot certify legal renewals. It instead constructs and evaluates a proxy: an *observable successor procurement*, a later BOAMP episode from the same buyer that is sufficiently similar to, and temporally plausible relative to, an earlier awarded digital procurement episode. Every downstream result – survival curves, hazard ratios, trend segments – is conditioned on this proxy, not on certified legal ground truth. The strongest defensible framing is: because legal renewal status is not directly observed, the study measures observable successor procurements under a precision-first linkage rule, then assesses how sensitive its descriptive conclusions are to that rule.

2 What The Pipeline Measures

Let i index an earlier awarded procurement episode and j index a later candidate episode. The object of interest is not a legal renewal certificate, because BOAMP notices do not provide

that ground truth consistently. Instead, the pipeline estimates whether the data show a later procurement episode that is credible enough to be treated as a successor:

$$Y_i = \begin{cases} 1, & \text{if one accepted observable successor is found before the cutoff,} \\ 0, & \text{otherwise.} \end{cases}$$

The event time is

$$\tau_i = C_{\hat{j}} - A_i,$$

where A_i is the award-date origin of episode i , $C_{\hat{j}}$ is the first-publication date of the accepted successor, and $\hat{j}$ is the selected candidate. If no accepted successor is found before 2025-12-31, the episode is right-censored.

3 End-to-End Pipeline

The implemented pipeline is:

BOAMP notices → standardised notices → procurement episodes
→ candidate pairs → linkage scoring → one successor or abstention → survival dataset.

Standardisation and feature engineering. The preprocessing stage standardises dates, buyer identifiers, buyer names, CPV codes, text fields, procedure information, framework flags, and duration fields where they are explicitly available. It does not impute missing duration or invent expected expiry dates. Buyer standardisation is deliberately conservative: municipal variants such as commune/ville/mairie can be harmonised, but intercommunal legal forms are preserved as distinct entities, and conflicting validated SIRENs block a buyer match. The current legal-form audit reports 0 accepted links with conflicting validated SIREN evidence and 0 accepted municipal/intercommunal mixes.

Episode reconstruction. BOAMP is notice-level data, but the analysis needs procurement-level events. Therefore notices belonging to the same procurement are reconstructed into an episode. This avoids treating a consultation notice, correction notice, award notice, or lot-level administrative notice as separate renewals.

Candidate generation. Candidate generation is intentionally broad and is not the final model. For an anchor episode with award date A_i and candidate publication date C_j , the candidate is exposed only if

$$A_i + 90 \leq C_j \leq A_i + 2920.$$

The lower bound removes very near follow-up notices and parallel administrative activity. The upper bound is approximately eight years; it keeps the candidate pool inclusive enough for long public-procurement cycles. Precision is then controlled by the selection stage rather than by a narrow time window. The current blocking rule generated 763,417 candidate pairs, with a median of 89 candidates per anchor.

4 Grand Ouest Regional Reference

Evaluation uses a regional reference sample drawn from the study region itself: 120 awarded digital procurement anchors stratified by CPV theme, buyer-identifier quality, and duration availability across Bretagne, Pays de la Loire, and Normandie, reviewed on 2026-08-11 against the notices and official BOAMP URLs of each candidate. The pilot split carries 16 anchors and 5,221 pair rows; the locked split carries 69 anchors and 20,917 pair rows.

Two anchor counts appear for the locked split and the difference is by construction, not a discrepancy: 72 anchors are evaluable at anchor level, of which 69 also have at least one exposed candidate pair and so appear in the pair-level table. The remaining 3 generated no candidate at all, which is a blocking-stage loss counted against recall rather than a scoring failure. Anchor-level metrics use 72; pair-level ROC and precision-recall curves use 69.

This reference replaced an earlier France-level benchmark whose labels were emitted by deterministic rules reading the same text, CPV, and date evidence the linkage methods consume. That construction made the comparison circular: a method could only score well by agreeing with a hand-written rule built from its own features, and the resulting numbers measured rule agreement rather than correctness. It has been removed from the repository in full – its data, its construction scripts, and its annotation tooling – so nothing here can descend from it. Its history remains in version control.

Five limitations bind everything computed from the regional reference. The labels were produced by a single LLM research pass over the supplied notices, their official URLs, and wider public sources, then spot-checked on a subset by the project owner rather than verified anchor-by-anchor; anchors outside that subset carry the model’s judgement as recorded, so this is a reference sample and not ground truth. The sources behind each individual label were not recorded, so a given anchor’s evidence trail cannot be fully reconstructed or independently re-executed. Negatives are corpus-relative: roughly 25 candidates per anchor were considered rather than the full pool, so a false-positive rate computed on them is conservative by construction and is a diagnostic on this sample rather than a population-wide rate. And while no linkage method existed or was consulted when the labels were made, judging whether later notice text continues an earlier need draws on the same text, CPV, and date evidence the text-ranking method scores, so the labels are method-independent without being fully evidence-independent. The sample is small enough that every point estimate needs its interval read beside it. And the anchors were re-resolved onto the current episode reconstruction through their BOAMP notice identifiers, because the review recorded episode identifiers from an earlier reconstruction; 8 anchors did not resolve to exactly one current episode and were dropped rather than guessed.

5 Linkage Algorithms

All methods operate on the same exposed candidate set. This is crucial: the primary method is not comparing text over the whole BOAMP universe. Buyer and time plausibility are imposed before text ranking.

M_A : deterministic evidence. Define B_{ij} as buyer-identity support, P_{ij} as CPV continuity, and T_{ij} as text similarity. The deterministic rule accepts a link only when strong buyer evidence is present, CPV continuity is positive, and a minimum text signal is present:

$$B_{ij} = 1, \quad P_{ij} > 0, \quad T_{ij} \geq t_A.$$

It is interpretable, but it loses recall when CPV or buyer identifiers are missing or noisy.

M_B : text ranking. For each episode, the text fields are converted to TF-IDF vectors x_i and x_j . Text similarity is cosine similarity:

$$T_{ij} = \cos(x_i, x_j) = \frac{x_i \cdot x_j}{\|x_i\| \|x_j\|}.$$

Within the same-buyer, plausible-time candidate set, the method selects

$$\hat{j}_i = \arg \max_j T_{ij},$$

and accepts the selected candidate only if

$$T_{ij\hat{i}} \geq 0.70.$$

This is the primary method because it is simple, reproducible, auditable, and best matches the precision-first objective.

M_C : weighted gated score. This method combines evidence components into a score:

$$S_{ij} = 0.50B_{ij} + 0.25T_{ij} + 0.20P_{ij} + 0.05G_{ij},$$

where G_{ij} is a timing-plausibility score. Missing components are handled by the implemented score-normalisation logic rather than imputation. The method then applies gates and ranks by S_{ij} . It is useful as a higher-recall contrast, but the validation results show a materially higher false-positive rate.

M_D : Fellegi–Sunter probabilistic linkage. Let γ_{ij} be the discretised comparison vector for buyer, text, CPV, and timing evidence. Fellegi–Sunter estimates the likelihood of observing γ_{ij} under a match class M and a non-match class U , then uses the log-likelihood ratio

$$w_{ij} = \log \frac{P(\gamma_{ij} | M)}{P(\gamma_{ij} | U)}.$$

The fitted model provides `fs_match_weight` and `fs_match_probability`. In the current data it does not outperform the simple text-ranking rule, likely because true successors are rare and same-buyer procurement activity contains many non-renewal lookalikes.

A removed arm. A duration-conditioned variant was built and evaluated during development, then removed from the repository in full. Reliable duration is missing for most of the cohort, so such a rule can differentiate itself only on a minority of episodes, and where the evidence does exist the observed data show many declared successors published well before the declared end date. The event-definition sensitivity framework is therefore the four threshold and method arms ($M_B@0.80$, $M_B@0.70$, $M_B@0.60$, $M_C@0.70$), which vary the decision rule along the dimension that actually moves the event set, together with the borderline-band check. The removed arm’s history remains in version control.

The separate descriptive comparison between declared contract duration and observed successor delay is a different thing and remains part of the evidence: it is a diagnostic about duration reliability, not a linkage algorithm.

6 Survival Modeling

The accepted-link decision from M_B defines the survival event. For episode i , τ_i is the time from award date A_i to the accepted successor’s publication date if $Y_i = 1$, or to the study cutoff (2025-12-31) if $Y_i = 0$ (administrative right-censoring). $Y_i = 0$ means no accepted observable successor was found before the cutoff; it is not proof of abandonment.

Kaplan–Meier. The non-parametric survivor function $S(t) = P(T > t)$ is estimated by [Kaplan and Meier \(1958\)](#)’s product-limit estimator. It is reported overall and stratified by CPV segment, which is the only stratification this study estimates; group differences use a multivariate log-rank test across those segments. Region and framework status enter the Cox model as covariates but are not estimated as separate Kaplan–Meier strata.

Cox proportional hazards. The semi-parametric model is

$$h(t | X) = h_0(t) \exp(\beta_1 X_1 + \cdots + \beta_p X_p),$$

with covariates selected for substantive relevance and data quality rather than automated search: CPV digital segment, buyer region, framework-agreement flag, validated-SIREN availability, and centered award year. The rule of one covariate per ten observed events (Van Belle et al., 2002) is respected with 544 events supporting 8 covariates. The proportional-hazards assumption is tested with Schoenfeld residuals (Grambsch and Therneau, 1994); violations are reported and interpreted descriptively rather than silently dropped or used to discard the model.

Parametric models. Exponential, Weibull, log-logistic, log-normal, and generalized-gamma models are compared by log-likelihood, AIC, and BIC and checked graphically against the Kaplan–Meier curve. Their role is to identify the best-fitting family and to provide the instrument any extrapolation past the observation window would require. They are *not* the source of the reported 12/24-month probabilities: every horizon quoted in this report falls inside the observed window, and the smooth families flatten the empirical renewal shoulder, so the operational conditional probabilities are read off the Kaplan–Meier estimator, which imposes no shape.

Temporal validation. The model is fit once on episodes awarded 2015–2021 and scored out of time without refitting. The primary evaluation window is 2022–2024, as specified by the internship guideline; 2022–2025 is carried as a sensitivity read that adds the shortest-follow-up award cohort. Harrell’s concordance index C is reported on each split to assess discrimination and out-of-time stability, not to target a specific value.

Evaluation strategy. The three checks applied to the linkage – scoring it against a reference subset, comparing linked with unlinked episodes, and re-running the analysis under alternative linkage rules – follow the guidance of [Harron et al. \(2017\)](#) for evaluating linkage quality in linked-data analyses. That source supports the strategy; it does not validate the precision, recall, or threshold reported here.

Borderline-link robustness. Threshold sensitivity swaps one event definition for another; a separate check asks how much rests on the anchors the frozen rule nearly classified the other way. Anchors whose best candidate scores within ± 0.05 of the 0.70 acceptance bar are dropped entirely – borderline acceptances and borderline abstentions alike – and the headline Kaplan–Meier levels and hazard ratios are recomputed on the remainder. The band is fixed a priori and is not searched over.

Linkage sensitivity. Because the event is linkage-conditioned, the same Kaplan–Meier and Cox analyses are repeated under the strict ($M_B@0.80$), main ($M_B@0.70$), looser ($M_B@0.60$), and high-recall contrast ($M_C@0.70$) event definitions. A conclusion is reported as robust only when it is stable in sign and approximate magnitude across these arms.

7 Trend And Change-Point Detection

Quarterly awarded-episode counts $N_{s,q}$ are built for the overall cohort and each CPV digital segment from 2015Q2 (the first complete quarter) through 2025Q4, including zero-count quarters.

Change-point detection (PELT). [Killick, Fearnhead and Eckley \(2012\)](#)’s PELT algorithm minimizes a penalized segmentation objective on the z-standardized series with penalty $\beta = \lambda \log(n)$; the central result uses $\lambda = 1$ with sensitivity at 0.5 and 2.0, and a break is called stable only if it lies within one quarter under all three penalties.

Stationarity. Augmented Dickey–Fuller (unit-root null) and KPSS (level-stationary null) tests are run on each segment’s series. The two tests have opposite null hypotheses and are read together, not individually.

Regime detection (HMM). A 3-state Gaussian hidden Markov model is fit on the quarter-over-quarter *change* in episode count – not the level – for the overall cohort and the two highest-volume CPV segments, so states describe typical period-over-period direction (decline, plateau, growth) rather than absolute activity level. The Viterbi-decoded state sequence gives the current-quarter regime and its posterior probability. This complements PELT (which finds discrete historical breaks) with a continuously updated regime read; the two are not required to agree, and disagreement is reported honestly rather than reconciled.

Recent direction. An OLS slope over the latest 12 quarters gives an `increasing/decreasing/stable_or_uncertain` label at exploratory $\alpha = 0.10$, uncorrected for multiple testing. None of these methods forecasts future values or identifies the cause of a break; causal attribution requires documentary or stakeholder evidence not available in BOAMP alone. Monetary and duration trend series are omitted because no canonical awarded-amount field is validated at episode grain, and 2025’s duration-field completeness jump is a measurement change, not a genuine shift in contract durations.

8 Technology Segmentation

The internship guide’s L2 deliverable specifies a supervised technology taxonomy: 300–500 manually annotated contracts across 8–12 classes, a TF-IDF+logistic-regression/SVM baseline evaluated by macro-F1 and confusion matrix, and an optional CamemBERT comparison. **This was not built inside the reproducible analytical branch.** Two independent annotators with a Cohen’s kappa agreement statistic are required for a defensible L2 corpus, and no second qualified annotator was available within this session’s scope; producing single-pass AI-assisted labels and calling their agreement "kappa" would repeat exactly the self-consistency problem this project has documented for the successor-linkage benchmark (§4) instead of avoiding it.

A finer technology-classification effort was carried out in parallel, outside this pipeline: `data/reference/technology_classification/` holds a 945-row export dated 2026-08-12 carrying a `Domaine` label per notice. It could not be integrated at the analytical freeze and is read by no stage of this pipeline. It covers current national opportunities rather than the 2015–2025 Grand Ouest study cohort, and it arrives without the training corpus, the annotation guidelines, or the validation artifacts that would let its labels be audited or applied historically. The present reproducible branch therefore retains CPV divisions as the primary technological segmentation. That is a statement about what could be integrated and verified here, not a judgement on the parallel work.

Every technology segment referenced in this report – cohort selection, Cox covariates, quarterly trend series – therefore uses CPV divisions 32 (telecommunications equipment), 35 (security), 48 (software), and 72 (IT services) as a reproducible substitute. This is a real scope reduction, not a disguised classifier: CPV divisions are official, zero-missingness EU categories under Regulation 213/2008, but they are coarser than a learned taxonomy and cannot distinguish sub-themes such as cloud versus on-premise infrastructure within a division. A future session with a second qualified annotator could complete L2 by reusing the blinded-review design recorded in `INDEPENDENT_LINK_REVIEW_PROTOCOL.md`; the scripts that once implemented it were removed with the retired France-level benchmark and remain recoverable from version control.

9 Dev Results

Method	Threshold	Accepted	Precision	Recall	FPR	Coverage
M_A_deterministic	70.000	3	0.667	0.400	0.091	0.188
M_B_text_ranking	70.000	2	0.500	0.200	0.000	0.125
M_C_weighted_gated	70.000	5	0.800	0.800	0.000	0.312
M_D_fellegi_sunter	65.000	2	0.000	0.000	0.091	0.125

Table 1: Unweighted dev metrics on all labelled current benchmark frames.

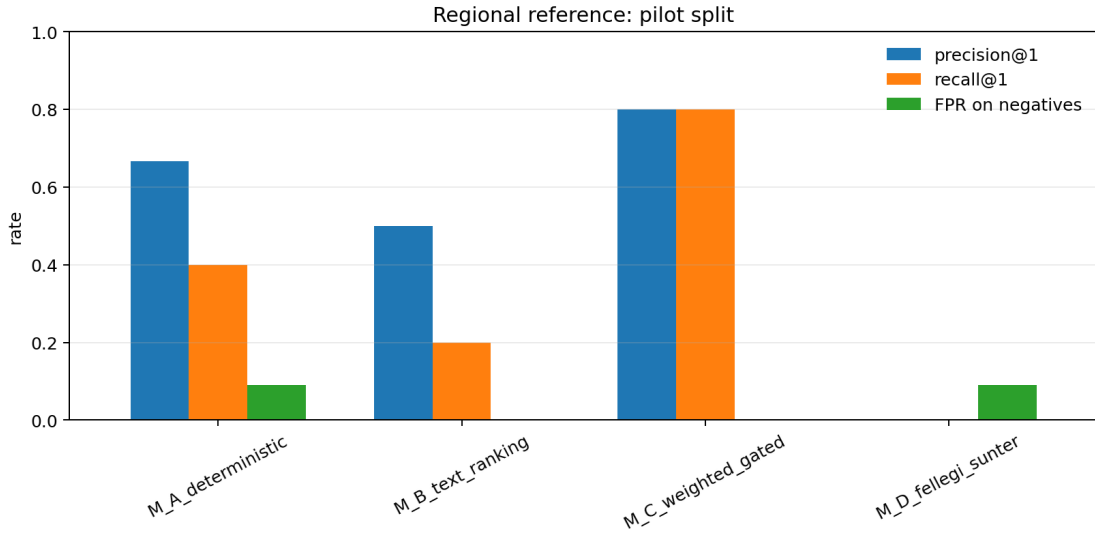


Figure 1: Dev split method comparison generated from the current benchmark evaluation JSON.

10 Internal Held-Out Reference Results

The table below is an internal method-comparison diagnostic. It must not be presented as external validation. The primary comparison is anchor-level exact-successor performance: each anchor can produce one accepted successor or abstain.

$$\text{precision} = \frac{TP}{TP + FP}, \quad \text{recall} = \frac{TP}{TP + FN}, \quad \text{FPR} = \frac{FP}{FP + TN}.$$

For this project, false-positive control is prioritised because false links fabricate survival events.

Method	Threshold	Accepted	Precision	Recall	FPR	Coverage
M_A_deterministic	70.000	15	0.533	0.444	0.130	0.208
M_B_text_ranking	70.000	8	0.875	0.389	0.000	0.111
M_C_weighted_gated	70.000	23	0.522	0.667	0.185	0.319
M_D_fellegi_sunter	65.000	5	0.200	0.056	0.018	0.069

Table 2: Unweighted held-out metrics on the locked split of the regional reference.

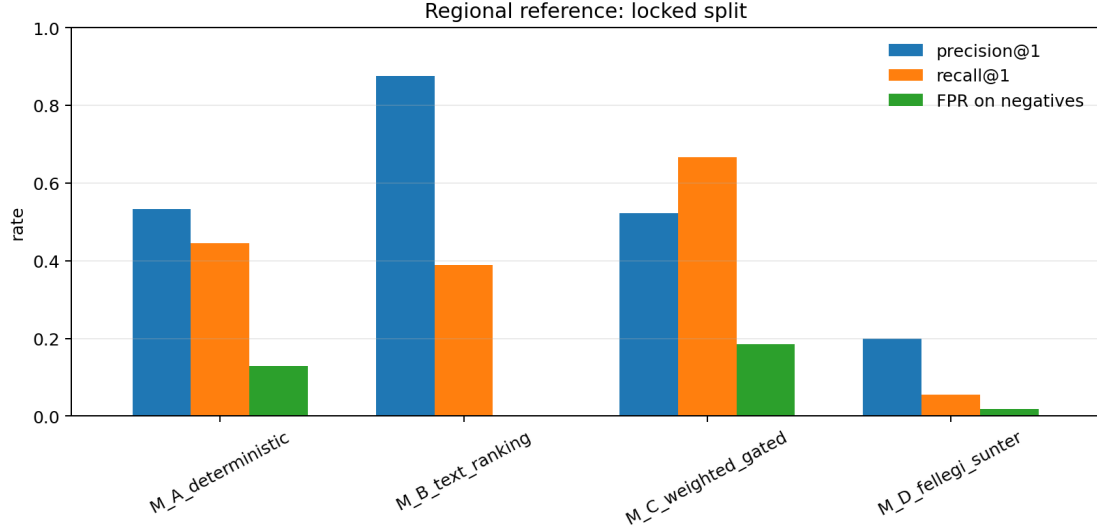


Figure 2: Held-out method comparison generated from the regional-reference evaluation JSON.

11 Quality Evidence And Interpretation

The held-out internal result supports retaining a conservative baseline for continued work; it does not establish final accuracy. M_B has only 8 accepted held-out links, so its precision estimate is necessarily sample-sensitive; one changed decision would move the number materially. Within this reference, the direction is coherent: M_B gives the best precision and the lowest false-positive rate among useful methods, while M_C shows the expected precision-recall trade-off.

The pair-level ROC and precision-recall curves should be read as score-ranking diagnostics, not as the final operating decision. The final pipeline decision is anchor-level: choose one successor or abstain. The ROC curve is stepped rather than smooth because the held-out reference contains a finite number of labelled pairs and many tied or near-tied scores.

The emphasis on precision-recall for rare positive decisions is supported by [Davis and Goadrich \(2006\)](#) and [Saito and Rehmsmeier \(2015\)](#). These sources justify the diagnostic choice; they do not validate the reference labels, numerical results, or selected threshold. Generic web illustrations are therefore treated as presentation aids only, not as academic evidence.

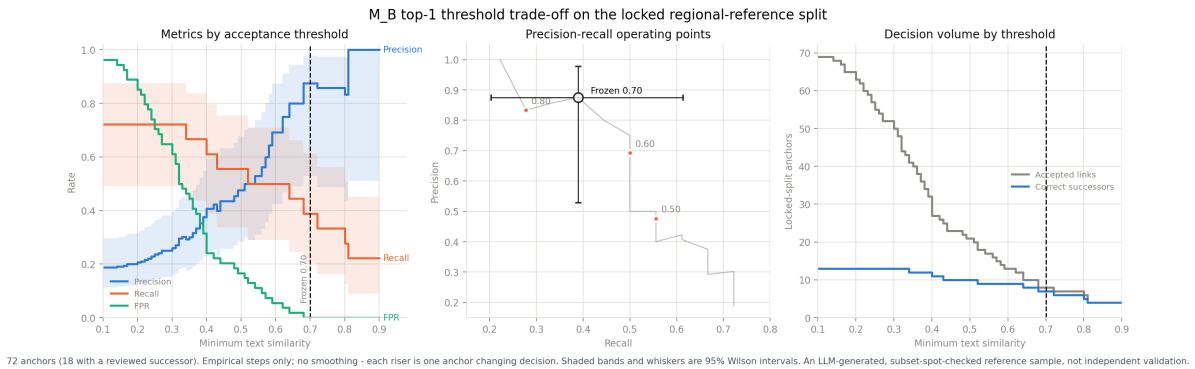


Figure 3: Project-specific, unsmoothed anchor-level threshold trade-off for M_B. The figure is empirical evidence from the locked split of the regional reference, not an idealised illustration.

On the locked split, 0.70 accepts 8 links of which 7 are correct (precision 0.875, recall 0.389, FPR 0.000), while 0.60 accepts 13 of which 9 are correct (precision 0.692, recall 0.500, FPR

0.056). On the pilot split the same two points give precision 0.500 and 0.667 respectively. The threshold was fixed before this reference was read, which is the only reason the locked split can be reported as held out; selecting one from these rows now would convert it into a tuning set. The lower threshold is therefore reported as a sensitivity arm rather than promoted after inspecting validation.

12 Model-Assisted Challenge Review

A separate blinded challenge review sampled 20 current accepted links, 20 high-similarity structural-negative candidates, and 20 buyer-declared relationships. Its provenance is model-assisted rather than independent human specialist review. Among the 20 accepted links, 14 were confirmed, 5 were rejected, and 1 was uncertain. Conservative reviewed precision was 0.700, with exact 95% interval $[0.457, 0.881]$. This diagnostic falls below the 0.80 point target and exposes useful failure modes. It does not establish independent human validation, but it is sufficient to justify keeping the current claim narrow and the lower threshold unpromoted.

13 Modeling Tables

The modeling-ready tables include strict, primary, broad, and non-match target columns, plus observable features. They now include the Fellegi–Sunter columns `fs_match_weight` and `fs_match_probability`, so modeling and evaluation use the same feature state.

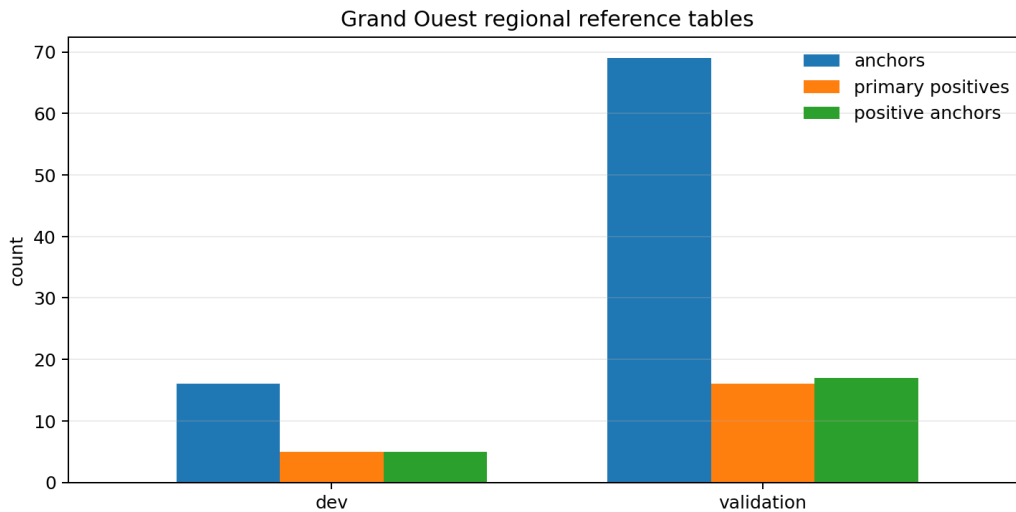


Figure 4: Current benchmark modeling table sizes and label support.

14 Survival Results

The survival dataset uses the main accepted links as events and all other episodes as right-censored observations at the study cutoff. The main variant contains 544 events and 3,256 censored observations (14.316% event rate). Segment event rates are 0.132 for CPV-32, 0.204 for CPV-35, 0.117 for CPV-48, and 0.143 for CPV-72. CPV-35 has the highest observed successor rate in the main arm.

Kaplan–Meier. The estimated probability of an observable successor is 4.621% by 12 months, 6.733% by 24 months, and 17.535% by 60 months. The Kaplan–Meier median survival time is

not reached: the curve never falls below 0.5 within the observation window, so no median is reported. This is distinct from the 31.82-month median delay *among linked events only*, which conditions on the event having occurred. A multivariate log-rank test across CPV segments gives statistic 23.45 ($p = 3.26e - 05$).

Cox model. The parsimonious model (8 covariates, 544 events) gives:

Covariate	HR	95% CI	p
framework_flag	1.751	[1.435, 2.136]	3.38e-08
has_validated_siren	1.082	[0.885, 1.323]	0.443
award_year_centered	1.107	[1.066, 1.149]	1.21e-07
digital_segment_CPV-35	1.553	[1.218, 1.981]	0.000383
digital_segment_CPV-48	0.828	[0.638, 1.073]	0.153
digital_segment_CPV-72	1.056	[0.850, 1.310]	0.624
buyer_region_Normandie	0.800	[0.640, 1.000]	0.0501
buyer_region_Pays de la Loire	1.003	[0.815, 1.234]	0.979

Table 3: Cox hazard ratios, main linkage arm. In-sample C-index 0.626.

Framework-agreement episodes and CPV-35 carry the largest hazard ratios among substantively interpretable covariates. These are descriptive associations with the observed hazard, not causal effects.

Covariate	Test statistic	p	PH assumption
award_year_centered	70.733	4.09e-17	violated
buyer_region_Normandie	0.608	0.436	ok
buyer_region_Pays de la Loire	2.455	0.117	ok
digital_segment_CPV-35	0.041	0.84	ok
digital_segment_CPV-48	0.671	0.413	ok
digital_segment_CPV-72	3.723	0.0537	ok
framework_flag	7.040	0.00797	violated
has_validated_siren	6.763	0.00931	violated

Table 4: Schoenfeld-residual proportional-hazards tests.

Proportional-hazards diagnostic. The assumption is rejected ($p < 0.05$) for `award_year_centered`, `framework_flag`, `has_validated_siren`. These coefficients are therefore reported as time-averaged associations rather than constant hazard ratios; stratification or time-interaction terms would be the next refinement if individualized prediction were required.

Temporal validation. Training on 2015-2021 (2,470 episodes, 392 events) and evaluating on the guideline-aligned 2022-2024 window (1,004 episodes, 107 events) gives C-index 0.606 in-sample versus 0.479 out-of-time. Extending the test window to 2022-2025 (1,330 episodes, 152 events), without refitting, gives 0.518. Both are close to the 0.5 chance line: individualized out-of-time discrimination is weak. The two windows also differ in follow-up, and [Uno et al. \(2011\)](#) show that the concordance statistic for right-censored data converges to a quantity depending on the censoring distribution, so the gap between the two figures should not be read as a change in discriminative ability. That is a result, not a prompt to retune – the model is retained as a descriptive risk-factor summary and nothing in the operational deliverable rests on it. Part of the gap is structural, because episodes awarded from 2022 onwards can contribute only short-gap events.

Borderline-link robustness. Dropping the 280 episodes whose best candidate scores within ± 0.05 of the acceptance threshold removes 133 events and leaves the direction of both headline hazard ratios unchanged (CPV-35 1.55 to 1.78; framework 1.75 to 1.62). The absolute Kaplan–Meier level falls, from 4.6% to 3.7% at 12 months, which is the mechanical consequence of removing borderline events. The comparative conclusions therefore do not rest on borderline linkage decisions, while absolute probabilities remain threshold-uncertain.

Model	Parameters	Log-likelihood	AIC	BIC
GeneralizedGamma	3	-3757.5	7521.0	7539.7
LogNormal	2	-3784.6	7573.2	7585.7
LogLogistic	2	-3807.3	7618.6	7631.1
Weibull	2	-3814.7	7633.4	7645.8
Exponential	1	-3829.7	7661.3	7667.6

Table 5: Parametric survival model comparison.

Parametric models. `GeneralizedGamma` has the lowest AIC and BIC and was checked graphically against the Kaplan–Meier curve. It is reported as the best-fitting family and as the instrument any extrapolation past the 2025-12-31 cutoff would use. It is *not* the source of the exported 12/24-month conditional successor probabilities (`survival_conditional_probabilities.csv`): those come from the Kaplan–Meier estimator, each with a 500-draw episode-bootstrap interval. The reason is visible in the fit itself – around age 36 months, just before the observed renewal shoulder, every parametric family including the selected one reports roughly 0.02–0.03 for the next twelve months where Kaplan–Meier reports near 0.075. Since every horizon this report quotes falls inside the observed window, the extrapolation that would justify accepting that smoothing is not needed. These are estimated probabilities of an *identifiable observable successor procurement*, not certified renewal probabilities, and no active Gigalis portfolio was available to score, so they cover the study cohort rather than a live deployment set.

Detectability and selection diagnostic. Comparing linked and censored episodes on standardized mean differences, the largest gap is `text_length_chars` (SMD 0.262). This indicates possible differential detectability across episode characteristics, not proof of causal linkage bias; it cannot be fully separated from genuine heterogeneity in renewal behaviour using BOAMP alone.

Linkage sensitivity. Event counts range from 296 to 1,332 across the four retained linkage arms, so absolute probabilities are linkage-sensitive by construction. Cox effects under the same four arms:

Covariate	Strict	Main	Looser	High-recall	Robustness
<code>award_year_centered</code>	1.017	1.107	1.136	1.101	Robust
<code>buyer_region_Normandie</code>	0.803	0.800	0.794	0.726	Mostly Robust
<code>buyer_region_Pays de la Loire</code>	0.977	1.003	1.023	0.891	Linkage Sensitive
<code>digital_segment_CPV-35</code>	1.792	1.553	1.488	1.484	Robust
<code>digital_segment_CPV-48</code>	0.652	0.828	0.807	0.885	Mostly Robust
<code>digital_segment_CPV-72</code>	1.108	1.056	1.036	1.557	Linkage Sensitive
<code>framework_flag</code>	1.281	1.751	1.953	1.623	Robust
<code>has_validated_siren</code>	1.193	1.082	1.107	1.650	Linkage Sensitive

Table 6: Cox hazard ratios across linkage-definition sensitivity arms.

Framework flag, CPV-35, and centered award year are the most robust associations; buyer-region and CPV-72 effects are linkage-sensitive and should not be over-interpreted.

15 Trend Results

Segment	Direction	Slope/qtr	p	Last stable PELT break	HMM regime
Overall	stable_or_uncertain	-0.11	0.921	–	growth
CPV-32	stable_or_uncertain	-0.01	0.989	2020Q2	growth
CPV-35	stable_or_uncertain	0.03	0.923	–	–
CPV-48	decreasing	-0.84	0.032	2024Q1	–
CPV-72	stable_or_uncertain	0.70	0.285	2021Q1	growth

Table 7: Current trend signal matrix: OLS 12-quarter slope, PELT breaks, and HMM current regime (Overall and top-2 segments only).

CPV-48 is the only segment with a statistically distinguishable 12-quarter decline at the exploratory $\alpha = 0.10$ level; the rest are **stable_or_uncertain** by this signal.

Operational reading. Each segment’s signals translate into a monitoring action, carried in the **business_recommendation** column of **trend_signal_matrix.csv** so that this text and the materialised table cannot drift apart. These are readings of descriptive evidence, not forecasts and not causal explanations: a PELT break dates a level shift without explaining it, and none of the statements below should be quoted as attributing a shift to policy, COVID, regulation, or technology.

- **Overall** (stable_or_uncertain): Maintain monitoring for Overall; no statistically clear recent direction at the pre-declared exploratory level. The HMM currently reads this series as ‘growth’, which describes recent quarter-over-quarter change and need not agree with the 12-quarter slope.
- **CPV-32** (stable_or_uncertain): Maintain monitoring for CPV-32; no statistically clear recent direction at the pre-declared exploratory level. A penalty-stable level shift is dated 2020Q2; treat it as a break candidate to be explained with documentary evidence, not as a demonstrated cause. The HMM currently reads this series as ‘growth’, which describes recent quarter-over-quarter change and need not agree with the 12-quarter slope.
- **CPV-35** (stable_or_uncertain): Maintain monitoring for CPV-35; no statistically clear recent direction at the pre-declared exploratory level.
- **CPV-48** (decreasing): Investigate the recent decline in CPV-48 before reducing or expanding procurement capacity; confirm whether it reflects demand, publication practice, or a routing change to another channel. A penalty-stable level shift is dated 2024Q1; treat it as a break candidate to be explained with documentary evidence, not as a demonstrated cause.
- **CPV-72** (stable_or_uncertain): Maintain monitoring for CPV-72; no statistically clear recent direction at the pre-declared exploratory level. A penalty-stable level shift is dated 2021Q1; treat it as a break candidate to be explained with documentary evidence, not as a demonstrated cause. The HMM currently reads this series as ‘growth’, which describes recent quarter-over-quarter change and need not agree with the 12-quarter slope.

Segment	ADF (H0: unit root)	KPSS (H0: level stationary)	Note
Overall	stat=-5.890, p=0.000	stat=0.268, p=0.100	
CPV-32	stat=-6.086, p=0.000	stat=0.808, p=0.010	
CPV-35	stat=-1.684, p=0.439	stat=0.444, p=0.058	
CPV-48	stat=-1.749, p=0.406	stat=0.545, p=0.032	
CPV-72	stat=-5.439, p=0.000	stat=0.611, p=0.022	

Table 8: Stationarity diagnostics per segment.

ADF and KPSS test opposite null hypotheses and are read jointly; disagreement between them means the series is not cleanly classified as stationary or non-stationary over this 43-quarter window.

The HMM regime column above reports each covered segment’s current-quarter regime and posterior probability, fit on the quarter-over-quarter change in episode count rather than its level. Because the model is fit on noisy, low-count quarterly data, the **plateau** state is a data-driven middle tier rather than a change centered exactly at zero, and it is read alongside **decline/growth** rather than as "no change." The HMM’s current-quarter regime and the 12-quarter OLS slope are complementary diagnostics computed over different windows; they are not required to agree, and this report does not adjust either to match the other.

16 Defensible Decision

The final project decision is to keep **M_B_text_ranking @ 0.70** as the frozen conservative observable-successor baseline and use the stricter ($M_B@0.80$), looser ($M_B@0.60$), and weighted-gated ($M_C@0.70$) variants as the required sensitivity analyses, together with the borderline-band check. The threshold is not claimed to be optimal; it is claimed to be pre-specified. It was fixed before the regional reference was consulted and has not been moved since, so the locked-split figures are held out rather than fitted. Moreover, the completed model-assisted review confirmed only 14 of 20 sampled production links at 0.70 conservatively, so lowering an unreviewed threshold would not support a stronger accuracy claim. This is a precision-first design. Low recall is accepted as an explicit trade-off, but the observed event rate is not a mathematical lower bound: missed successors push it downward while residual false links can push it upward. [Doidge and Harron \(2019\)](#) make the general point that missed and false links misclassify in opposite directions, which is why no one-sided bound follows from a linkage-conditioned event rate. It is therefore a linkage-conditioned indicator whose absolute value must be read with the strict, looser, and weighted-gated sensitivity results and the borderline-band check. Independent specialist review is required only before claiming externally validated precision or promoting a new threshold, not to complete this linkage-conditioned descriptive study. If that review shows weaker precision, the claim should be narrowed further rather than forcing a more complex model.

17 Limitations And Robustness

- BOAMP does not consistently encode legal renewal status; accepted links are observable successor procurements, not legal renewal proof.
- The reference labels were generated by a single LLM research pass and spot-checked on a subset by the project owner, not verified anchor-by-anchor and not reviewed by an independent specialist panel. Its negatives are corpus-relative: roughly 25 candidates per anchor were considered, so the reported false-positive rate is conservative by construction rather than a population-wide rate.

- The held-out reference is small: 18 positive anchors and 8 accepted M_B links. The selected threshold should not be over-tuned and its accuracy remains provisional.
- Buyer standardisation is improved but remains an important risk area. The legal-form audit is retained to catch name-only and cross-legal-form cases.
- Missing duration and expiry fields are not imputed. This avoids creating false timing certainty.
- The 3-state HMM regime labels are fit on noisy, short (43-quarter) count series; state-count and window choices are not exhaustively validated, and a ‘plateau’ label is a relative middle tier rather than a change centered at zero.
- Technology segmentation uses CPV divisions, not a learned taxonomy (§8); this is a real scope reduction, disclosed rather than concealed.

18 Perspectives

18.1 Causal Inference (Outline Only)

The internship guide frames a distinct causal question, separate from this report’s descriptive linkage/survival/trend results: does a Gigalis pooled framework agreement actually change member purchasing behaviour (amounts, frequency, segments), or does Gigalis simply capture demand that would have existed anyway? **This question is not answered here, and no causal estimate is computed in this repository.** Answering it requires data this project’s BOAMP-only corpus does not contain: which buyers are Gigalis members, and when each joined. Attempting a substitute causal claim from BOAMP alone – for example treating CPV-segment membership or region as a proxy treatment – would confound procurement domain and geography with Gigalis adoption, and this project has consistently avoided exactly this kind of unsupported inferential leap elsewhere (survival hazard ratios are reported as associations, not effects, for the same reason).

If Gigalis-internal membership and adoption-date data were joined to this corpus, the natural identification strategy is a **staggered-adoption difference-in-differences** design ([Callaway and Sant’Anna, 2021](#)), because members join the central purchasing body at different calendar dates rather than simultaneously – the specific setting that motivates that estimator over classical two-period DiD. The parallel-trends assumption (comparable purchasing trajectories absent Gigalis membership) would need to be assessed with pre-adoption trend plots per member cohort. Propensity-score matching ([Angrist and Pischke, 2009](#)) is a secondary candidate if adoption is not sharply timed but depends on observable buyer characteristics (size, budget, sector); regression discontinuity would apply only if a hard eligibility threshold (e.g. a population cutoff) determines Gigalis access, which is not currently known to be the case. Any of these designs should state its identifying assumptions as an explicit causal graph ([Pearl, Glymour and Jewell, 2016](#)) before estimation, not after.

This section is a design outline for future work conditional on Gigalis supplying membership data, not a promise that such data will become available within this internship.

19 Recommended Reporting Position

The technical report should defend the project as a reproducible measurement pipeline, not as a black-box prediction system. The strongest statement is:

Because legal renewal status is not directly observed in BOAMP, the study estimates observable successor procurements. Candidate generation restricts comparison to the

same buyer and a plausible future interval; the selected method then accepts only the strongest textually similar candidate above a fixed threshold. This prioritises precision over recall, which is appropriate because false positive links would create artificial survival events.

20 Current Source Files

- data/processed/boamp/linkage_evaluation_dev.json
- data/processed/boamp/linkage_evaluation_validation.json
- data/processed/boamp/regional_benchmark/modeling/modeling_summary.json
- data/processed/boamp/regional_benchmark/regional_benchmark_manifest.json
- data/reference/regional_link_benchmark/
- data/processed/boamp/survival_dataset_summary.json
- data/processed/boamp/linkage_candidates_summary.json
- data/processed/boamp/survival_analysis_summary.json
- data/processed/boamp/survival_cox_results.csv
- data/processed/boamp/survival_ph_diagnostics.csv
- data/processed/boamp/survival_parametric_comparison.csv
- data/processed/boamp/survival_cox_linkage_sensitivity.csv
- data/processed/boamp/trend_analysis_summary.json
- data/processed/boamp/trend_signal_matrix.csv
- DATA_QUALITY_REPORT.md
- TREND_ANALYSIS_REPORT.md
- SURVIVAL_ANALYSIS_REPORT.md
- PROJECT_WORK_PROTOCOL.md
- REVIEW_AUDIT_RESULTS.md
- EXECUTIVE_SUMMARY.md

21 Methodological References

The source and identity definitions follow the official BOAMP and INSEE documentation. CPV interpretation follows Commission Regulation (EC) No 213/2008. TF-IDF cosine similarity follows the standard vector-space definition. Record linkage, survival estimation, proportional-hazards diagnostics, and change-point detection are supported by the original Fellegi-Sunter (1969), Davis-Goadrich (2006), Saito-Rehmsmeier (2015), Kaplan-Meier (1958), Cox (1972), Grambsch-Therneau (1994), and PELT (Killick et al., 2012) methods. Regime detection follows the Markov-switching framework of Hamilton (1989). The causal-inference outline in §18.1 draws on Angrist and Pischke (2009), Pearl, Glymour and Jewell (2016), Athey and Imbens (2017),

and Callaway and Sant’Anna (2021). None of these sources validates this project’s own labels, numerical results, or thresholds; they justify method choices only. Full URLs and the specific design implications are recorded in `METHODOLOGICAL_REFERENCES.md`.

- [Official BOAMP API \(DILA\)](#).
- [INSEE SIREN definition](#) and [INSEE SIRET definition](#).
- [Commission Regulation \(EC\) No 213/2008 on CPV](#) and [Directive 2014/24/EU, Article 33](#).
- [Cosine similarity documentation](#).
- [Davis–Goadrich \(2006\)](#) and [Saito–Rehmsmeier \(2015\)](#) on ROC and precision-recall evaluation for skewed binary decisions.
- [Fellegi–Sunter \(1969\)](#), [Kaplan–Meier \(1958\)](#), and [Cox \(1972\)](#).
- [Grambsch–Therneau \(1994\)](#) and [Killick et al. \(2012\)](#).
- [Hamilton \(1989\)](#) on Markov-switching regime models, the framework underlying the trend HMM.
- [Angrist and Pischke \(2009\)](#), [Pearl, Glymour and Jewell \(2016\)](#), [Athey and Imbens \(2017\)](#), and [Callaway and Sant’Anna \(2021\)](#) on causal identification, cited only to support the outlined-not-executed causal-inference perspective.